



# **US Flight Delay Performance Dashboard Dimensional Modelling and Visual Analytics Approach**

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and Imaging**

# Overview of the Data Dataset: US Flight Delays

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## Flight Operational Records

- Airline identifier
- Origin and destination airport
- Year, Month, Day
- Scheduled and actual departure and arrival times
- Arrival and departure delay
- Cancellation and diversion flags
- Delay causes such as Weather, Airline, Air System, Security, Late Aircraft



## Analysis Objective

To analyze operational performance and delay patterns across airlines and airports.

## Data Transformations Performed

Preparing the dataset for dimensional modelling and analytical reporting

These transformations were necessary to convert raw operational data into analytical format, support star schema dimensional modelling, enable segmentation through slicers, improve performance insight, and ensure aggregation accuracy.



### Data Cleaning

- Replaced null delay cause values with 0 to prevent aggregation errors
- Standardised column data types
- Removed invalid or inconsistent records

### Time Standardisation

- Converted HHMM numeric time format into proper Time datatype
- Created FlightDate column from Year, Month, Day fields
- Derived Month Name, Quarter, and Day of Week attributes

### Delay Engineering

- Created Delay Category classification: On Time ( $\leq 0$  minutes), Minor Delay (1 to 15 minutes), Major Delay ( $> 15$  minutes)
- Created OnTimeFlag binary indicator
- Created Total Delay metric by summing delay causes

### Distance Segmentation

- Created Distance Band classification: Short Haul ( $\leq 500$  miles), Medium Haul (501 to 1500 miles), Long Haul ( $> 1500$  miles)
- Enabled comparative performance analysis across route types

# Data Modelling Approach Star Schema Implementation

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## Fact Table: **Fact\_Flights**

Contains measures including Arrival Delay, Departure Delay, Distance, Air Time, and Cancellation Flag



## Dimension Tables

- DimDate - temporal attributes
- Airline - airline descriptive data
- Airport - airport descriptive data



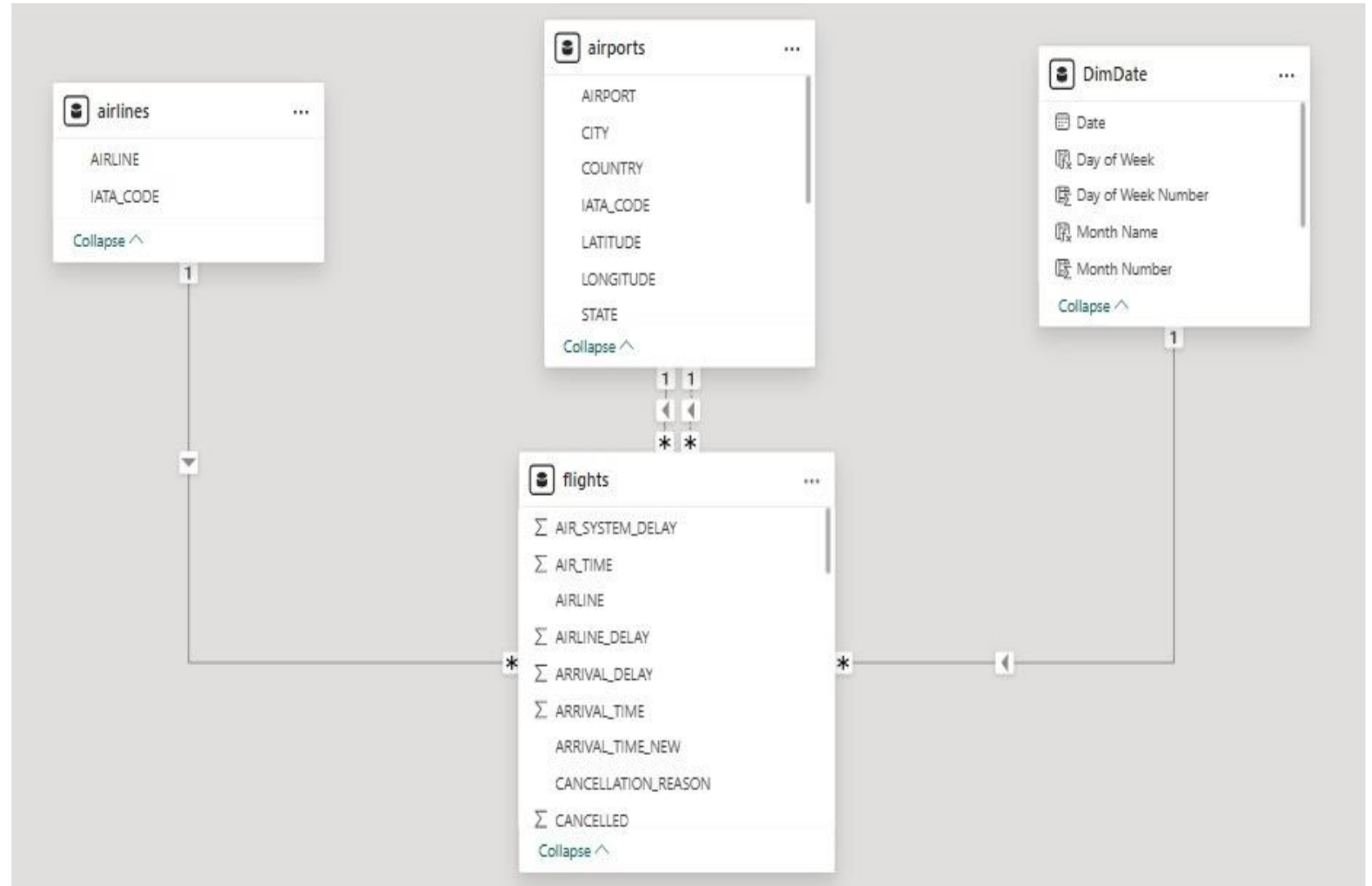
## Why Star Schema

- Simpler structure
- Optimized query performance
- Clear separation between measures and descriptive attributes
- Ideal for dashboard analytics

## Star Schema Design

Fact\_Flights stores transactional flight data. Dimensions store descriptive attributes used for slicing and filtering. This structure supports efficient filtering through slicers.

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# What, Why and How



## What

We analyzed flight delay performance across airlines, airports, and time dimensions.



## Why

To identify operational inefficiencies and delay causes that impact airline performance.



## How

- Cleaned and transformed raw data
- Built dimensional model
- Created DAX measures
- Designed interactive dashboard

# Visual Encoding Principles Applied

These principles improve interpretability and decision support.



## Position

Used in bar and line charts for comparing averages across categories



## Color

- Green for On Time flights
- Orange for Minor Delay events
- Red for Major Delay incidents



## Size

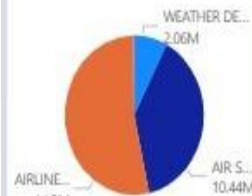
Map Airplane icon represents total flights volume



## Consistency

Uniform scales and minimal clutter for readability

DELAY CAUSES



| ORIGIN_AIRPORT | Major Delay | Minor Delay | On Time | Unknown | Total  |
|----------------|-------------|-------------|---------|---------|--------|
| ATL            | 29057       | 59818       | 228634  | 3330    | 32083  |
| ORD            | 33171       | 48025       | 163273  | 9330    | 25379  |
| DFW            | 25082       | 43836       | 139593  | 6904    | 21541  |
| DEN            | 18563       | 38389       | 112646  | 2653    | 17225  |
| LAX            | 15944       | 40396       | 112342  | 2670    | 17135  |
| IAH            | 16228       | 29947       | 84195   | 2603    | 13297  |
| PHX            | 12260       | 31621       | 87280   | 1263    | 13242  |
| SFO            | 11905       | 27308       | 89315   | 2517    | 13104  |
| LAS            | 10199       | 26243       | 79040   | 1244    | 11672  |
| MSP            | 9879        | 19871       | 73514   | 1062    | 10432  |
| SEA            | 10286       | 25689       | 66633   | 721     | 10332  |
| Total          | 480751      | 1049192     | 3627112 | 105071  | 526212 |

Total Flights by CITY





## **Key Visualisations**

Dashboard  
components  
supporting  
performance analysis

### **KPI Cards**

- High-level metrics summarising overall performance for quick monitoring.
- Total Flights: 5M
- On Time Rate: 68.9%
- Average arrival delay: -2.06
- Cancellation rate: 1.71%

### **Line Chart**

- Average Arrival Delay by Month - tracks temporal patterns and seasonality in delays across the year.

### **Bar Chart**

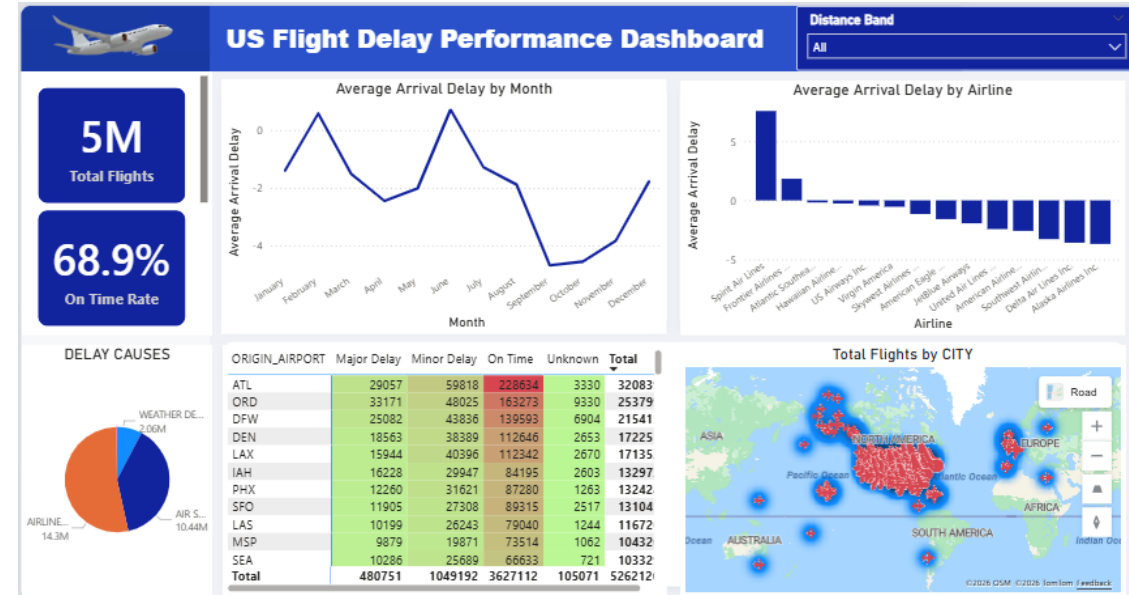
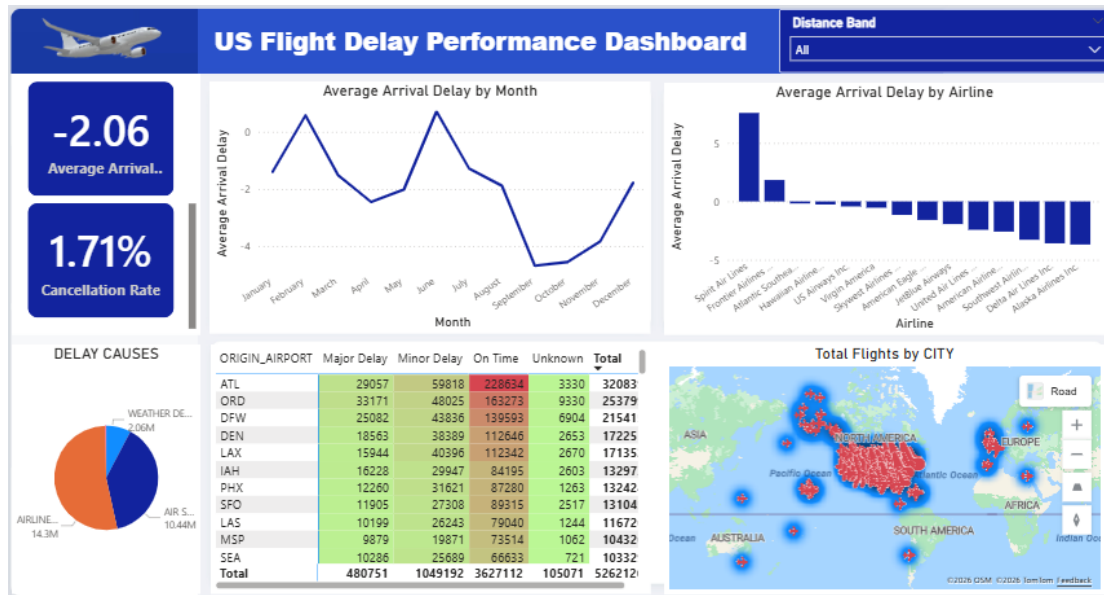
- Average Arrival Delay by Airline - compares carrier performance to highlight outliers and benchmarks.

### **Pie Chart**

- Distribution of Delay Causes - shows proportionate contribution of root causes such as weather, technical, and ATC.

### **Geographic Map**

- Flight volume by City - visualises spatial distribution and highlights major hubs and regional traffic patterns.



# Key Insights

## Findings from the dashboard analysis

- Arrival delays fluctuate significantly across months, with noticeable performance dips around late summer and early fall.
- September and October show the most negative arrival delay averages compared to earlier months.
- Several airlines maintain near zero or negative average arrival delays, indicating stronger schedule reliability.
- Major delays are concentrated among a smaller subset of airports rather than being evenly distributed.
- Airports such as ATL, ORD, DFW, and DEN contribute the highest total delay volumes due to high traffic throughput.
- High traffic airports show larger total delay counts, suggesting congestion driven inefficiencies.
- Minor delays significantly outnumber major delays across most airports, indicating short operational disruptions are more common than severe ones.
- Short haul flights demonstrate slightly better on time performance compared to medium and long haul routes, suggesting operational complexity increases with distance.



### 68.9% On Time Rate

Overall on-time performance stands at 68.9%, indicating moderate reliability with room for operational improvement.



### October Peak Delays

October shows highest negative delay performance, indicating strong seasonal operational disruption.



### Airline Variability

Spirit Airlines and Frontier Airlines show significantly higher average arrival delays compared to other carriers, highlighting performance variability across airlines.



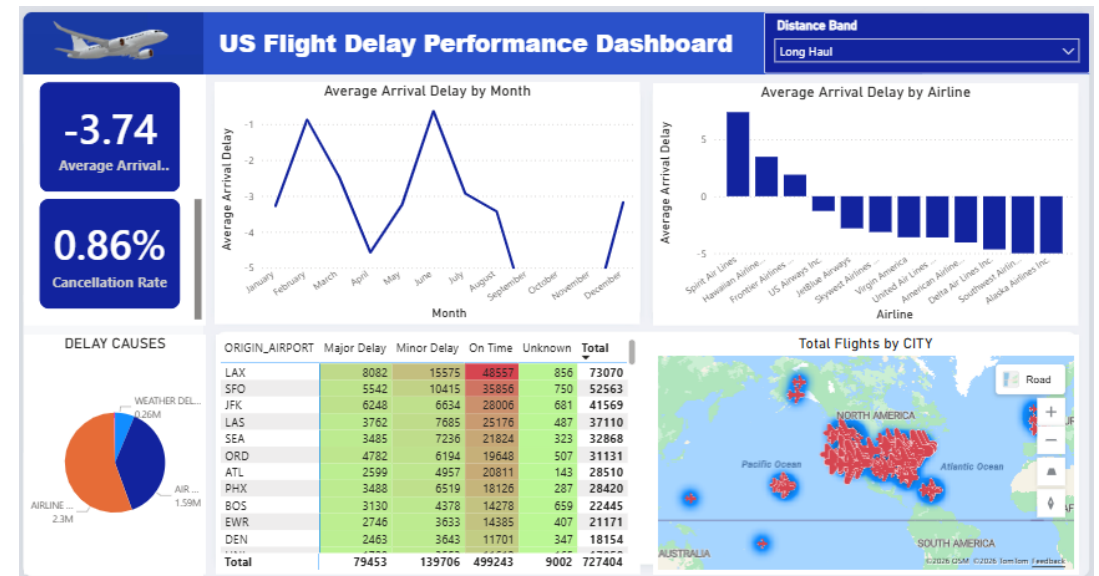
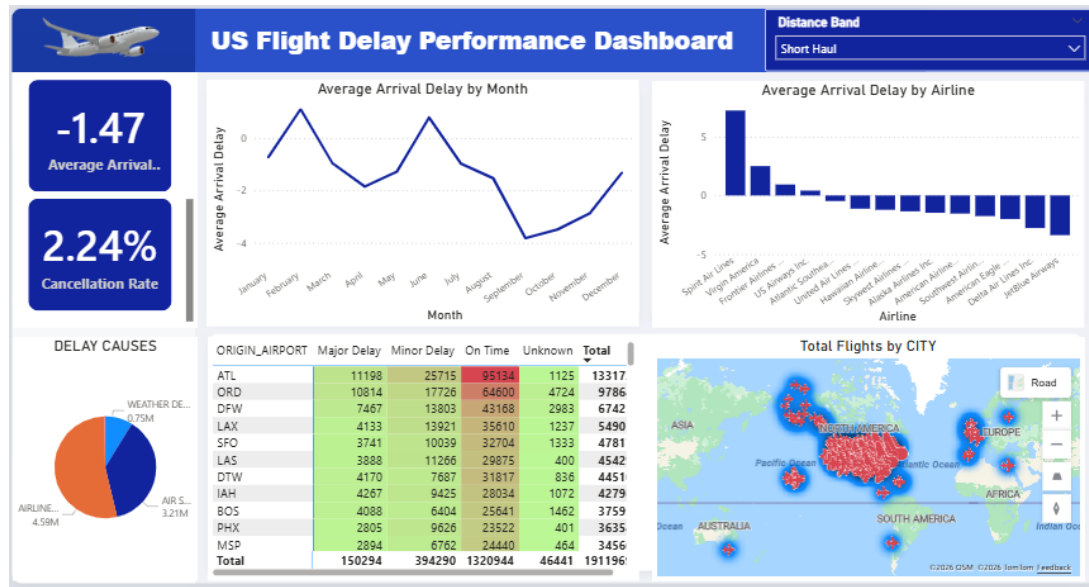
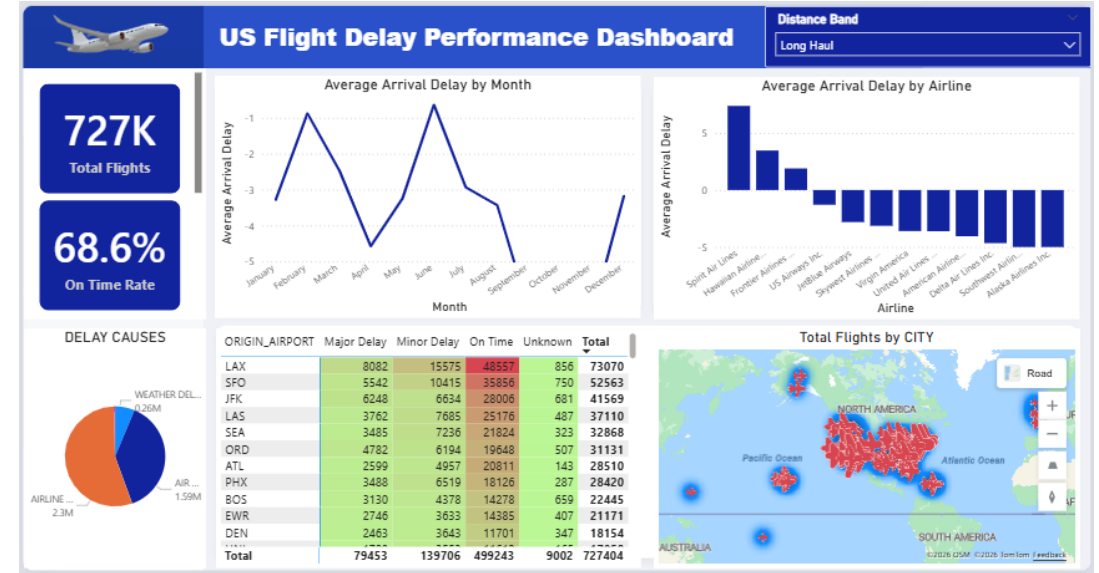
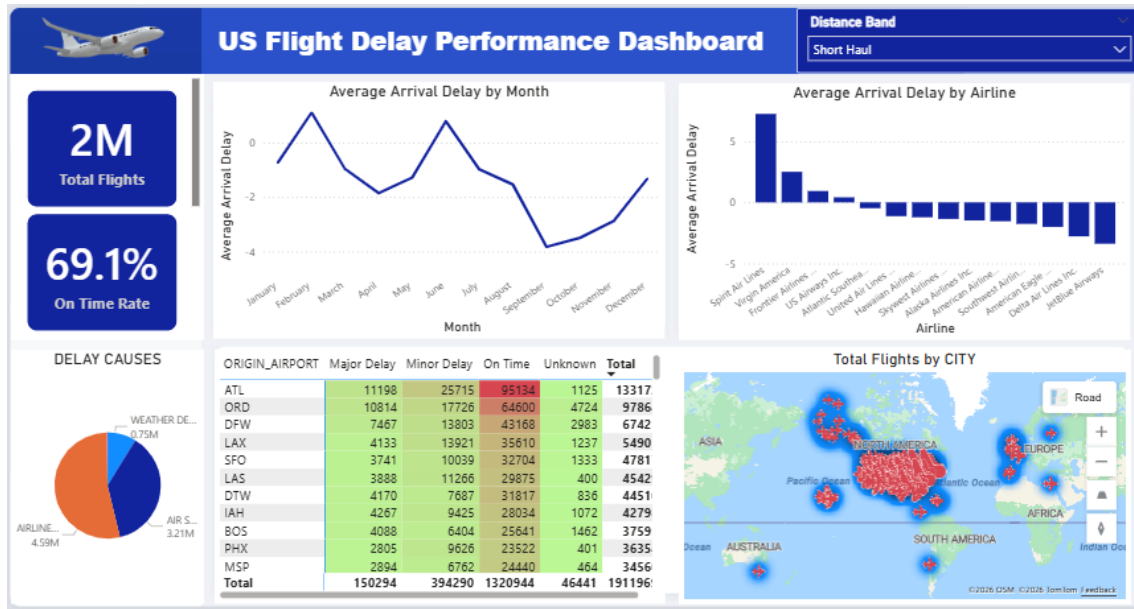
### Hub Congestion

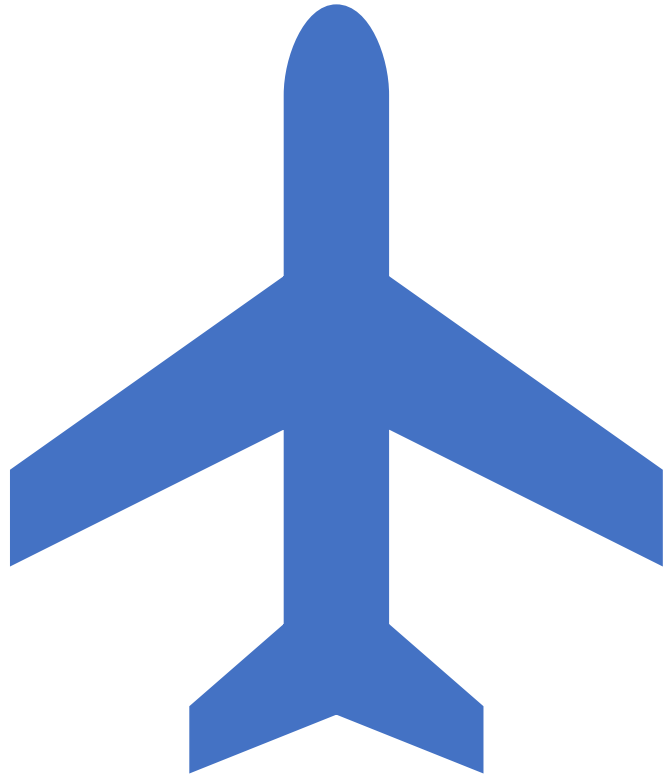
Major hub airports such as ATL and ORD contribute disproportionately high delay volumes due to traffic density and operational complexity.



### Delay Causes Dominance

Airline delays and Air System delays dominate overall delay causes, suggesting operational and air traffic management factors are primary contributors.





# Conclusion

The Star Schema enabled efficient performance analysis. The dashboard provides:

- Airline performance comparison
- Airport congestion insight
- Seasonal delay patterns
- Delay cause breakdown

Dimensional modelling combined with visual analytics improves operational monitoring and decision making.